

Chapter 1

INTRODUCTION TO NLP

Google Classroom code:

zw4lwctl

Overview: 3hrs

1. Introduction to NLP
2. Main components of NLP
3. Phases in NLP
4. Challenges in NLP
5. Ambiguity in NLP
6. Applications of NLP

Natural Language Processing

- Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that helps computers understand, interpret, and respond to human language — just like humans do.
- In simple words:
NLP teaches computers to **read, listen, and talk** using our natural languages (like English, Nepali, Hindi, etc.).

Analogy:

- Imagine you have a friend from another planet who doesn't speak your language.

You have to teach them:

- What words mean,
- How sentences are formed,
- What different tones or emotions mean.
- That's what NLP does — it teaches computers to understand and use human language correctly.

How NLP Works (Simple Idea)

- When you talk or type:
 - Your words are converted into a form the computer can understand (numbers or tokens).
 - The computer analyzes meanings, grammar, and context.
 - It gives a suitable **response or action**.
- For example:

If you type “What’s the weather like today?”, the computer recognizes:

 - “weather” → topic,
 - “today” → time,

and then gives you the correct weather report.

NLP for language translation:

Tout ce que vous produisez pour credit dans ce cours doit etre votre propre travail. Vous pouvez parler avec les autres etudiants (et les professeurs) de votre approche du probleme, mais ensuite vous devez rsoudre le prob leme par vous-meme. Ce n'est pas seulement la facon la plus ethique d'apprendre le contenu de cette classe, mais aussi la plus efficace.



All that you produce for credit in this course must be your own work. You can speak with the other stu dents (and the professors) about your approach about the problem, but then you must solve the same problem by you. It is not only the most ethical way to learn the contents from this class, but also most effective.

Information Extraction

Firm XYZ is a full service advertising agency specializing in direct and interactive marketing. Located in Bigtown CA, Firm XYZ is looking for an Assistant Account Manager to help manage and coordinate interactive marketing initiatives for a marquee automotive account. Experience in online marketing, automotive and/or the advertising field is a plus. Assistant Account Manager Responsibilities Ensures smooth implementation of programs and initiatives Helps manage the delivery of projects and key client deliverables . . . Compensation: \$50,000-\$80,000 Hiring Organization: Firm XYZ

INDUSTRY	Advertising
POSITION	Assistant Account Manager
LOCATION	Bigtown, CA.
COMPANY	Firm XYZ
SALARY	\$50,000-\$80,000

Named Entity Recognition

The screenshot displays a Named Entity Recognition (NER) interface. At the top, a blue header bar contains a legend with the following labels and their corresponding single-letter codes: Person (p, blue), Loc (l, yellow), Org (o, black), Event (e, green), Date (d, red), and Other (z, purple). Below the header, a text snippet is shown with several entities highlighted in colored boxes and marked with an asterisk (*). The entities and their labels are: 'Barack Hussein Obama II' (Person, blue), 'August 4, 1961' (Date, red), 'American' (Other, purple), 'the United States' (Loc, yellow), 'January 20, 2009' (Date, red), 'January 20, 2017' (Date, red), 'Democratic Party' (Org, black), 'African American' (Other, purple), 'United States Senator' (Other, purple), 'Illinois' (Loc, yellow), and 'Illinois State Senate' (Org, black).

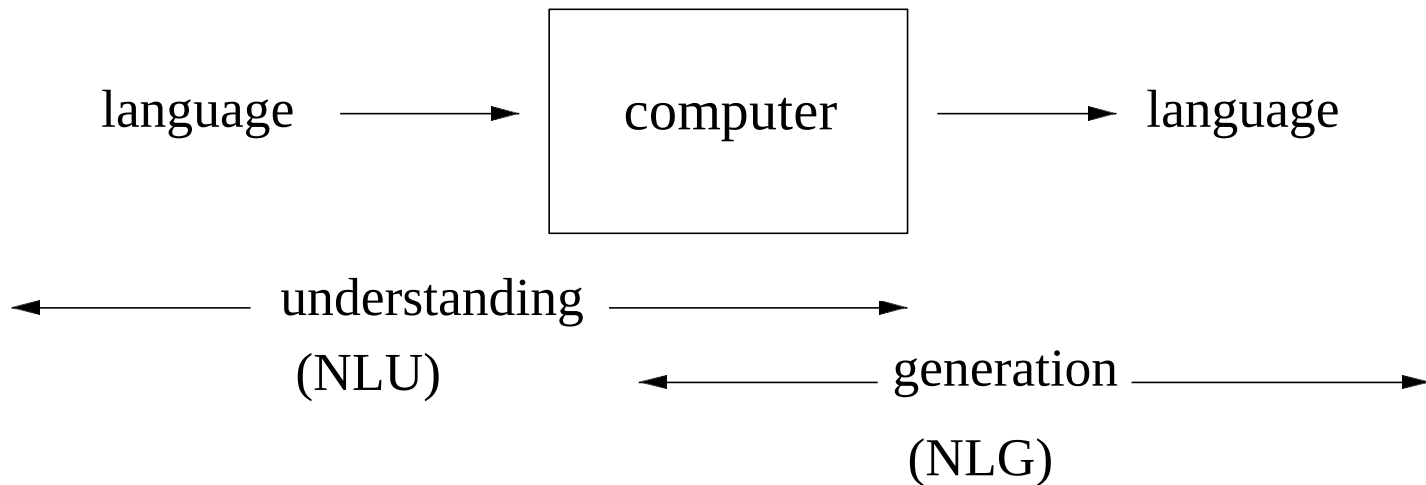
- The model can find different entities that are present in the text, such as persons, dates, organizations, and locations.

History and Evolution of NLP

Era	Approach	Characteristics	Examples
1950–1980s	Rule-based NLP	Manually written grammar rules; symbolic AI	ELIZA chatbot (1966)
1990–2010	Statistical NLP	Data-driven models using probabilities	Hidden Markov Models, POS tagging
2010–2018	Machine Learning / Deep Learning	Neural networks, embeddings, RNNs	Word2Vec, LSTM models
2018–Present	Transformers & Large Language Models	Attention mechanism, transfer learning	BERT, GPT, T5, ChatGPT

Main Components of NLP

- NLP = NLU (understanding) + NLG (responding)
- Like a conversation between humans, NLP first listens and understands, then thinks and replies.



1. Natural Language Understanding (NLU)

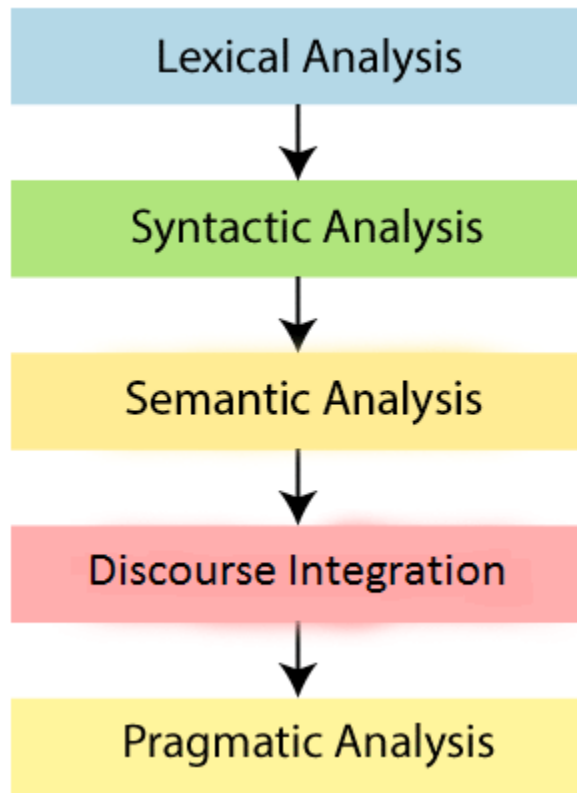
- This part helps the computer understand human language — what people mean when they speak or write.
- It includes:
 - a) Lexical Analysis
 - b) Syntactic Analysis (Parsing)
 - c) Semantic Analysis
 - d) Pragmatic Analysis
 - e) Discourse Integration

2. Natural Language Generation (NLG)

- Once the computer understands the input, it must generate a human-like response — for example, when ChatGPT answers you.
- It includes:
 - a) Content Planning**
 - Deciding what information to say.
 - Example: Weather app decides to say: “It will rain today.”
 - b) Sentence Planning**
 - Organizing that information into sentences.
 - Example: Turning data like {temp=25°C, rain=yes} into “It’s 25 degrees and raining.”
 - c) Text Realization**
 - Converting structured sentences into **natural human language** that sounds smooth.
 - Example: “It’s raining outside. Don’t forget your umbrella!”

Phases in NLP

- NLP processes human language in five main phases, starting from words and grammar to meaning and intention — just like how humans understand language step by step.



1. Lexical Analysis (Word-Level Analysis)

- This phase breaks the whole text into smaller parts like words or tokens and checks their structure and meaning.
- Like cutting a big sentence into small building blocks (words) before understanding it.
- **Activities include:**
 - Tokenization (splitting text into words)
 - Removing punctuation and stopwords
 - Identifying parts of speech (noun, verb, adjective, etc.)
- **Example:**
 - Sentence → “I love data science.”
 - Tokens → [I], [love], [data], [science]
 - Parts of Speech → Pronoun, Verb, Noun, Noun

2. Syntactic Analysis (Syntax or Grammar Checking)

- This phase checks how words are arranged in a sentence — whether the sentence follows correct grammar rules.
- The computer builds a “parse tree” to show how words relate grammatically.
- Example:
 - “He plays football.” : Correct
 - “Plays he football.” : Not Correct

3. Semantic Analysis (Meaning Analysis)

- This phase gives meaning to the sentence. It understands what each word and sentence means in context.
- Example:
 - “Ram went to the bank.”
 - Does “bank” mean *river bank* or *money bank*?
- Semantic analysis finds the correct one using context.

4. Discourse Integration

- It connects the current sentence with the previous sentences to make sense of the whole conversation or paragraph.
- Example:
 - “Aarushi bought a laptop. She loves it.”
 - Here, “she” refers to Aarushi — the system must link them correctly.

5. Pragmatic Analysis

- This phase interprets the intended meaning of a sentence in a real-world context — tone, situation, and purpose.
- Example:
 - “Can you open the window?”
 - Not a question about ability — it’s a polite request.

Challenges in NLP

Challenge	Description	Example
Ambiguity	Same word or sentence has multiple meanings	"I saw the man with a telescope." (Who has the telescope?)
Context Understanding	Words depend on context	"He is cool." → could mean calm or fashionable
Sarcasm Detection	Difficult for algorithms to detect tone	"Oh great, another rainy day!"
Multilinguality	Models must handle multiple languages and dialects	English vs Nepali grammar differences
Data Bias and Ethics	Training data may include societal bias	Biased sentiment classification models

Ambiguity in NLP

NLP is difficult because Ambiguity and Uncertainty exist in the language.

There are the following three ambiguity -

1. Lexical Ambiguity

- Lexical Ambiguity exists in the presence of **two or more possible meanings** of the sentence within a single word.
- **Example:** Maanashi is looking for a **match**.
- In the above example, the word match refers to that either **Maanashi is looking for a partner** or **Maanashi is looking for a match**. (Cricket or other match)

2. Syntactic Ambiguity

- Syntactic Ambiguity exists in the presence of **two or more possible meanings within the sentence**.
- **Example:** **I saw the girl with the binocular.**
- In the above example, did I have the binoculars? Or did the girl have the binoculars?

3. Referential Ambiguity

- Referential Ambiguity exists **when you are referring to something using the pronoun**.
- **Example:** **Gita went to Sunita. She said, "I am hungry."**
- In the above sentence, you do not know that who is hungry, either **Gita or Sunita**.

Advantages of NLP

1. **Language Understanding:** NLP helps computers understand human language for natural interactions.
2. **Information Extraction:** NLP extracts structured information from unstructured text, automating data analysis tasks.
3. **Language Translation:** NLP enables automatic translation between languages, facilitating cross-language communication.
4. **Sentiment Analysis:** NLP analyzes emotions in text, useful for social media monitoring and market research.
5. **Text Generation:** NLP generates human-like text, aiding content creation and personalized messaging.
6. **Efficiency and Automation:** NLP automates language-related tasks, saving time and resources.
7. **Insights and Decision Making:** NLP provides meaningful insights from text data, aiding data-driven decision making.
8. **Multilingual and Cross-Cultural Applications:** NLP supports multiple languages, fostering global communication.
9. **Accessibility:** NLP technologies make digital information accessible to people with disabilities.
10. **Personalization:** NLP understands individual preferences, improving personalized experiences.

Disadvantages of NLP

1. **Ambiguity and Context:** NLP **struggles with language ambiguity** and understanding context accurately.
2. **Cultural and Linguistic Variations:** NLP models may not generalize well across different languages, dialects, and cultural contexts.
3. **Data Availability and Quality:** Obtaining high-quality labeled data for training NLP models can be challenging, leading to limited resources and potential biases.
4. **Lack of Common Sense and World Knowledge:** NLP systems may struggle with common sense reasoning and lack general world knowledge.
5. **Ethical and Bias Concerns:** NLP models can perpetuate biases present in the training data, raising ethical concerns and potential unfair treatment.
6. **Language Complexity:** NLP **faces challenges in handling complex grammar, syntax, and semantics** in various languages.
7. **Contextual Limitations:** NLP models may have difficulty maintaining long-term context, impacting accurate interpretation and responses.
8. **Computational Resources:** NLP models can be computationally intensive, requiring substantial resources for training and inference.
9. **Constant Evolution:** NLP is a rapidly evolving field, requiring continuous updates and adaptation to changing language patterns and user needs.

Major Applications of NLP

- **Search Engines:**
NLP helps understand query intent.
Example: Google interprets “best hotels near me” using semantic search.
- **Chatbots and Virtual Assistants:**
Siri, Alexa, ChatGPT use NLP for dialogue and intent recognition.
- **Machine Translation:**
Tools like Google Translate or DeepL convert languages.
- **Text Summarization:**
Extracts main ideas from large documents.
- **Sentiment Analysis:**
Determines public opinion (e.g., “This product is amazing!” → Positive).
- **Speech Recognition:**
Converts spoken language into text (used in transcription apps).

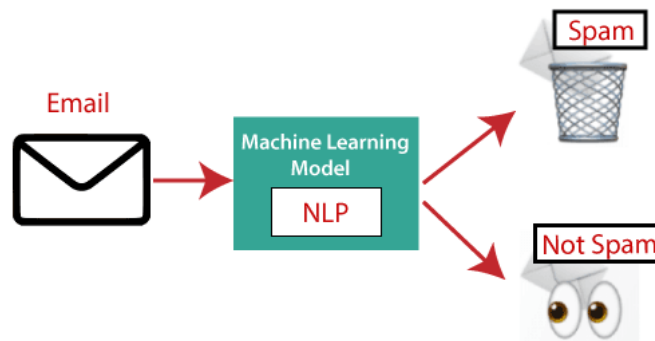
1. Question Answering

- Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.



2. Spam detection

- is used to detect unwanted e-mails getting to a user's inbox.



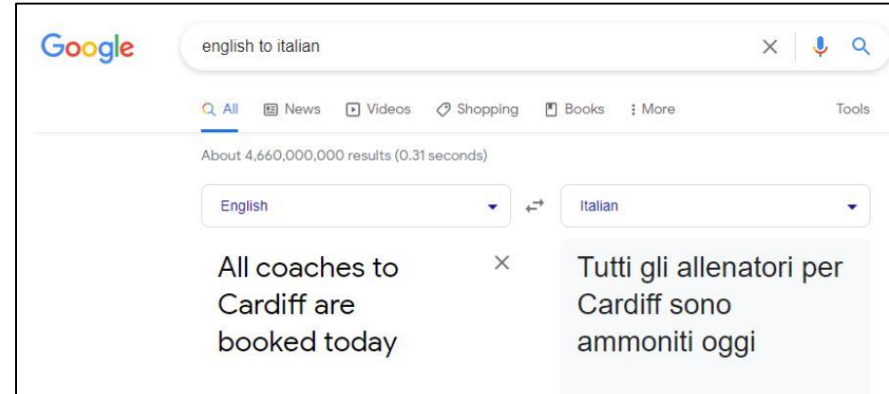
3. Sentiment Analysis

- Sentiment Analysis is also known as **opinion mining**.
- It is used on the web to analyze the attitude, behavior, and emotional state of the sender.
- This application is implemented through a combination of NLP and statistics by assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)



4. Machine Translation

- Machine translation is used to translate text or speech from one natural language to another natural language.
- **Example:** Google Translator



5. Spelling correction

- Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction



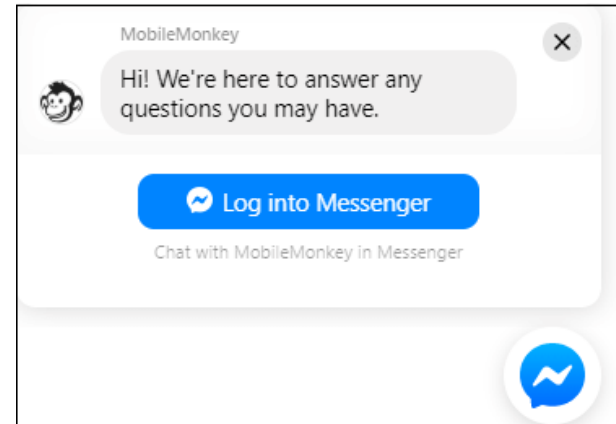
6. Speech Recognition

- Speech recognition is used for converting spoken words into text.
- It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.



7. Chatbot

- Implementing the Chatbot is one of the important applications of NLP.
- It is used by many companies to provide the customer's chat services.



8. Information Retrieval

- Information retrieval involves returning a set of documents in response to a user query: Internet search engines are a form of IR.
- Information extraction involves trying to discover specific information from a set of documents.
- However, one change from classical IR is that Internet search now uses techniques that rank documents according to how many links there are to them (e.g., Google's PageRank) as well as the presence of search terms.
- NLP (Natural Language Processing) plays a crucial role in information retrieval by enhancing the search and retrieval process of relevant information from large text collections.

information retrieval



NLP libraries

- **Scikit-learn:** It provides a wide range of algorithms for building machine learning models in Python.
- **Natural language Toolkit (NLTK):** NLTK is a complete toolkit for all NLP techniques.
- **Pattern:** It is a web mining module for NLP and machine learning.
- **TextBlob:** It provides an easy interface to learn basic NLP tasks like sentiment analysis, noun phrase extraction, or pos-tagging.
- **Quepy:** Quepy is used to transform natural language questions into queries in a database query language.
- **SpaCy:** SpaCy is an open-source NLP library which is used for Data Extraction, Data Analysis, Sentiment Analysis, and Text Summarization.
- **Gensim:** Gensim works with large datasets and processes data stream

A Multi-disciplinary field : NLP

- **NLP is multidisciplinary** because it combines the *language knowledge of linguistics*, the *logic of mathematics*, the *power of computer science*, the *intelligence of AI*, and the *insight of human cognition*.
- **Main Disciplines Involved**
 - a) **Linguistics** – gives knowledge about grammar, syntax, and meaning of language.
Example: Understanding that “bank” can mean money bank or river bank.
 - b) **Computer Science** – provides algorithms, programming, and data processing tools.
Example: Chatbots and search engines use computer programs to process text.
 - c) **Mathematics & Statistics** – used for probability, pattern recognition, and modeling.
Example: Predicting the next word in a sentence (language models).
 - d) **Artificial Intelligence & Machine Learning** – helps systems learn from data.
Example: Google Translate improves by learning from millions of sentences.
 - e) **Cognitive Science** – explains how humans think and understand language.
Example: Emotion or sentiment detection in text.

Assignment:

- Tokenization
- Stemming and Lemmatization
- WordNet
- Named Entity Recognition

End of chapter 1